



## CONTACT

📍 Isfahan, Iran  
📞 +98 914 040 9147  
✉ mohammadsoroushrabiei@gmail.com  
🐙 github.com/MohammadSoroushRabiei  
🌐 linkedin.com/in/mohammad-soroush-rabiei-0138a6368

## TECHNICAL SKILLS

### Programming

C Embedded C C++ Python

### MCUs

STM32 (F4/F7) ESP32 / ESP32-S3  
ESP8266

### Frameworks & Tools

STM32CubeMX/IDE Keil MDK ESP-IDF  
Git Doxygen

### Protocols & Peripherals

UART I<sup>2</sup>C RS-485 Modbus RTU  
MQTT DMA Timer input capture

### Debug & Test

ST-LINK Logic analyzer  
Hardware bring-up Simulators

### RTOS

FreeRTOS

### AI / ML

PyTorch OpenCV (in progress) ESP-DL  
Familiarity with machine learning and deep learning concepts

## LANGUAGES

🇮🇷 Persian native  
🇬🇧 English upper-intermediate (B2)  
🇨🇳 Chinese elementary (≈HSK 2)

# Mohammad Soroush Rabiei

## Embedded Systems Developer

## SUMMARY

Embedded systems developer with industrial experience in data-center infrastructure management (DCIM). My work and interests span IoT and connected devices, edge AI, and building smart products — mostly at the boundary of hardware and software.

## EXPERIENCE



### Embedded Systems Developer

Kian Pardaz Nagh-e Jahan · Isfahan, Iran

Jul 2026 – Present

- ▶ Researched and selected temperature/humidity sensors for the DCIM product.
- ▶ STM32 driver development :
  - For the DHT22 temp/hum sensor ( Non-Blocking State Machine + Timer Input Capture + DMA )  
🔗 <https://github.com/MohammadSoroushRabiei/STM32-DHT22-Driver>
  - For the Modbus RTU master over the RS-485 communication channel  
🔗 <https://github.com/MohammadSoroushRabiei/STM32-Modbus-RTU>



### Embedded Systems Intern

Behbood Ertebat Chehelsotoon Co. (Behamad) · Isfahan, Iran

Jun 2025 – Sep 2025

- ▶ Developed STM32 firmware for a smart-city (municipal IoT) device with LoRaWAN and a SIM800 GSM module.
- ▶ Implemented MQTT on STM32 to connect a smart thermostat to the platform.

## PROJECTS

### Smart Home — Self-Hosted Edge-AI Platform · B.Sc. final project

🔗 [github.com/MohammadSoroushRabiei/Smart-Home](https://github.com/MohammadSoroushRabiei/Smart-Home)

- ▶ A cloud-free smart home on ESP32-S3: on-device face recognition (ESP-DL), a self-learning on-chip behavior agent, an LVGL touch dashboard and a fully local MQTT/Home Assistant stack.

### IoT Smart Room · IoT course project

🔗 [github.com/MohammadSoroushRabiei/IoT\\_SmartRoom\\_ESP8266](https://github.com/MohammadSoroushRabiei/IoT_SmartRoom_ESP8266)

- ▶ Remote room control and monitoring (lights, fan, temperature) on ESP8266 with Blynk.

## EDUCATION & CERTIFICATES



### B.Sc. in Computer Engineering

Intelligent Systems · Isfahan University of Technology

2020 – 2026



### Diploma, Mathematics & Physics

NODET (SAMPAD) High School, Isfahan

2018



### Certificate of Participation — IoT 2025

The 9th International Conference on Internet of Things and Its Application · University of Isfahan

Oct 2025